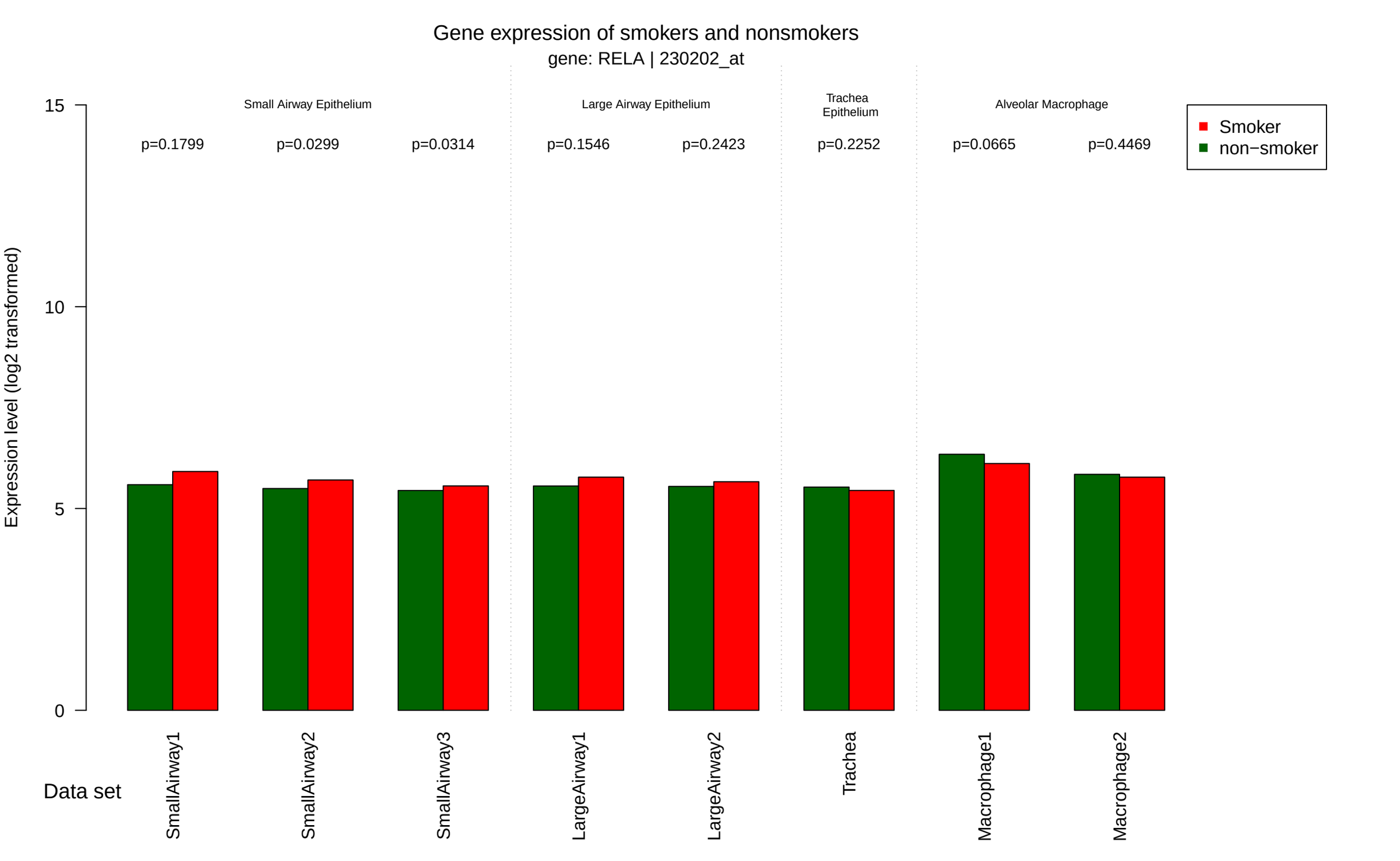
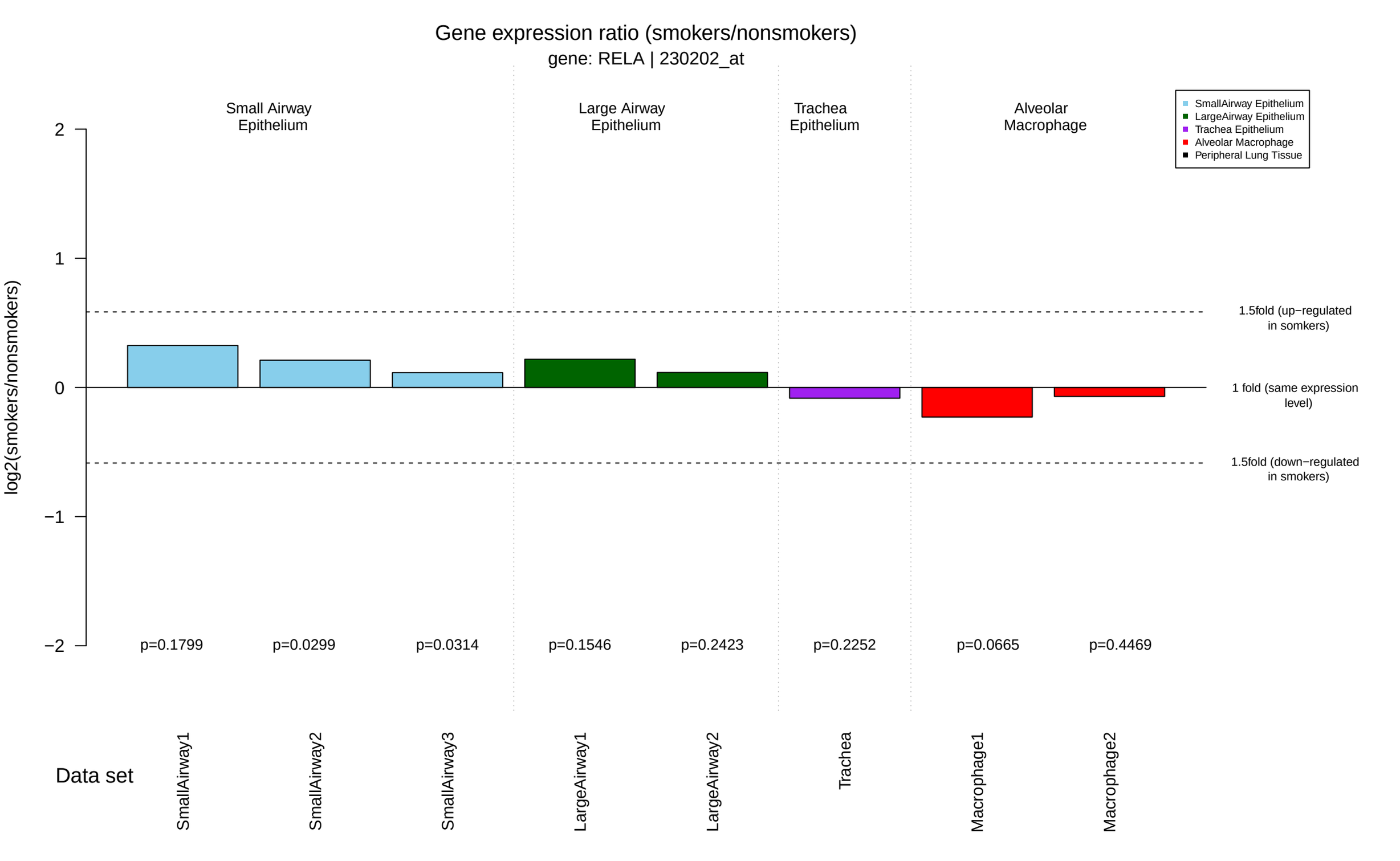
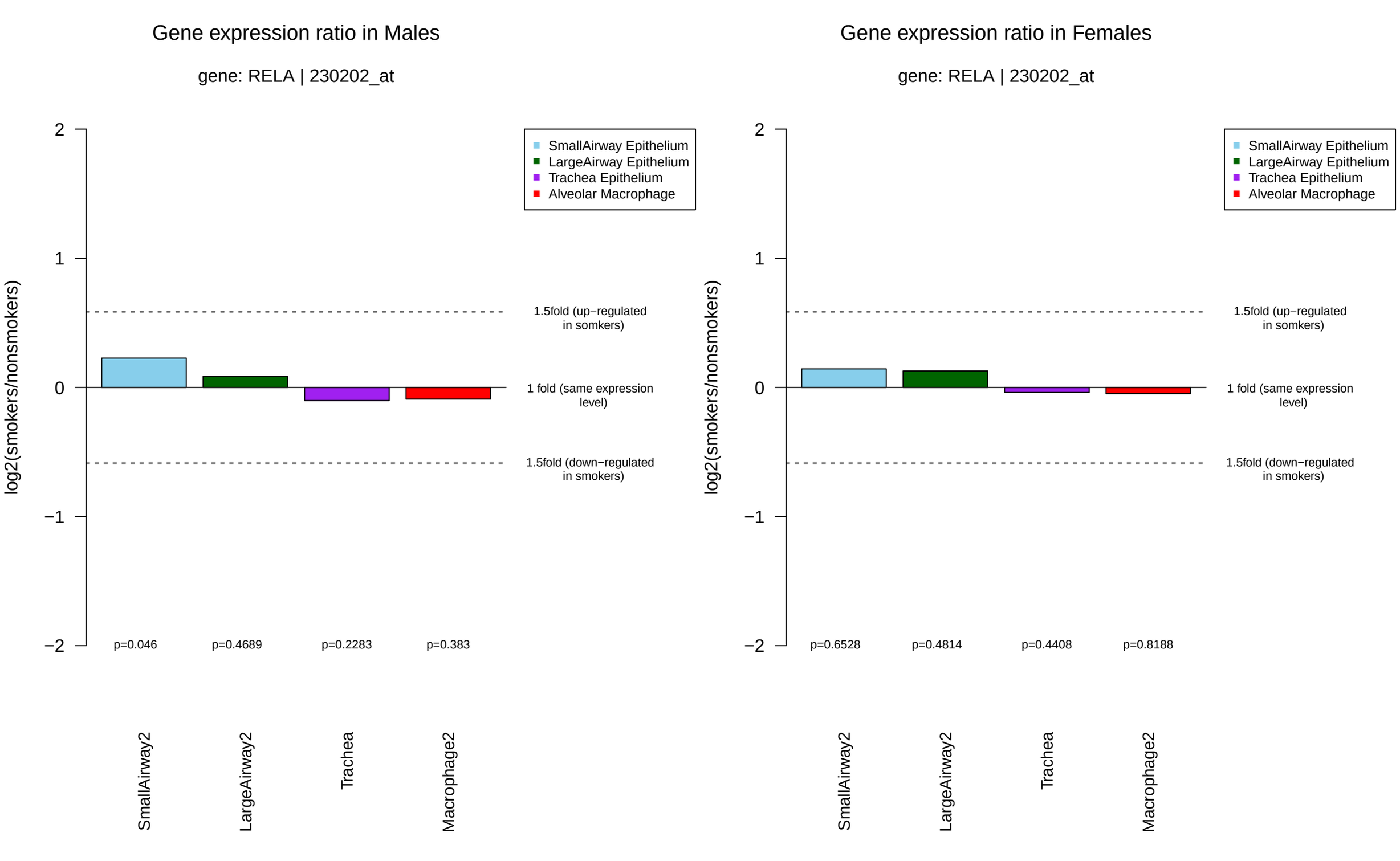
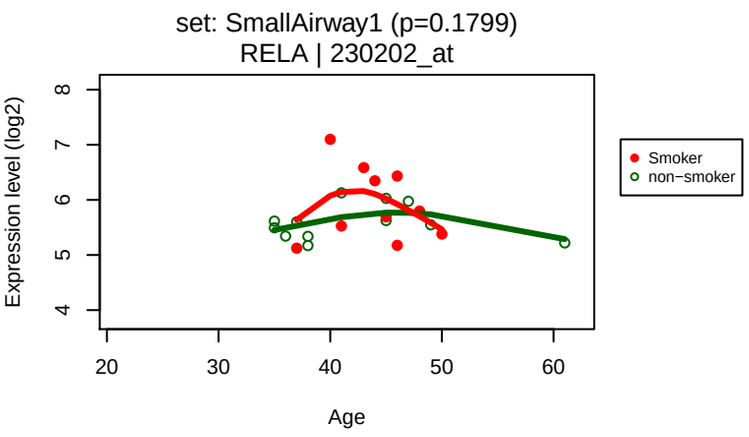
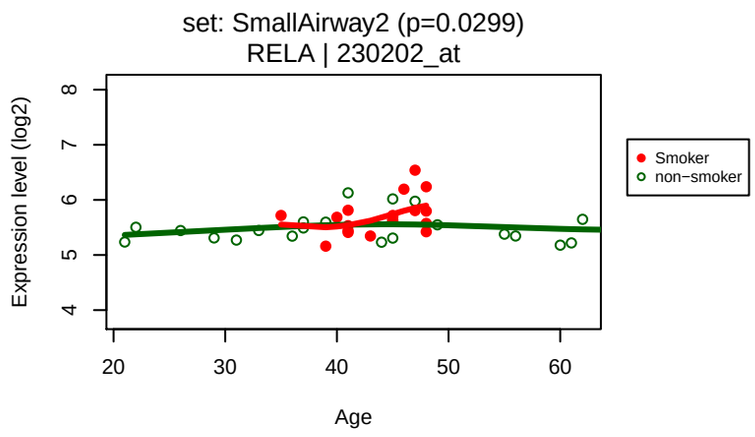
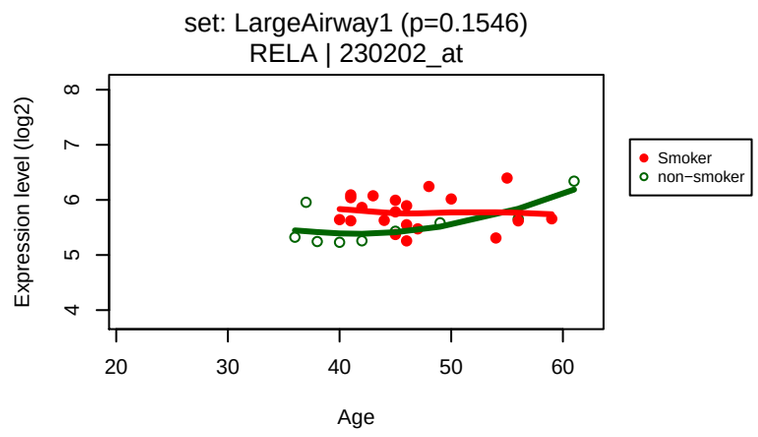

Gene expression of smokers and nonsmokers
gene: RELA | 230202_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1799
p=0.0299
p=0.0314
p=0.1546
p=0.2423
p=0.2252
p=0.0665
p=0.4469
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

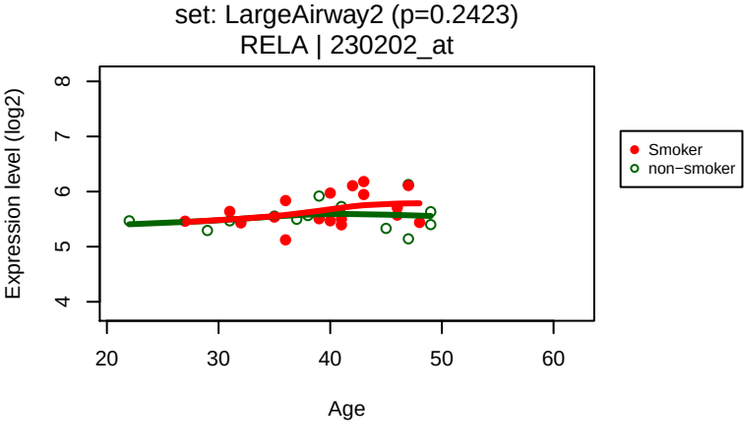

Gene expression ratio (smokers/nonsmokers)
gene: RELA | 230202_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1799
p=0.0299
p=0.0314
p=0.1546
p=0.2423
p=0.2252
p=0.0665
p=0.4469
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

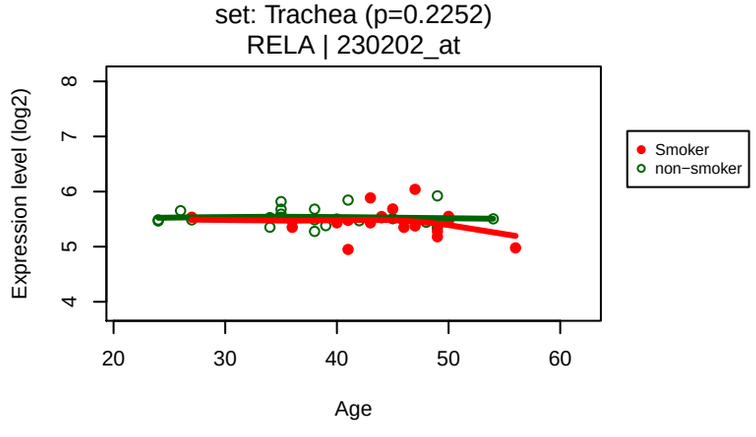

Gene expression ratio in Males
gene: RELA | 230202_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.046
p=0.4689
p=0.2283
p=0.383
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RELA | 230202_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6528
p=0.4814
p=0.4408
p=0.8188
SmallAirway2
LargeAirway2
Trachea
Macrophage2

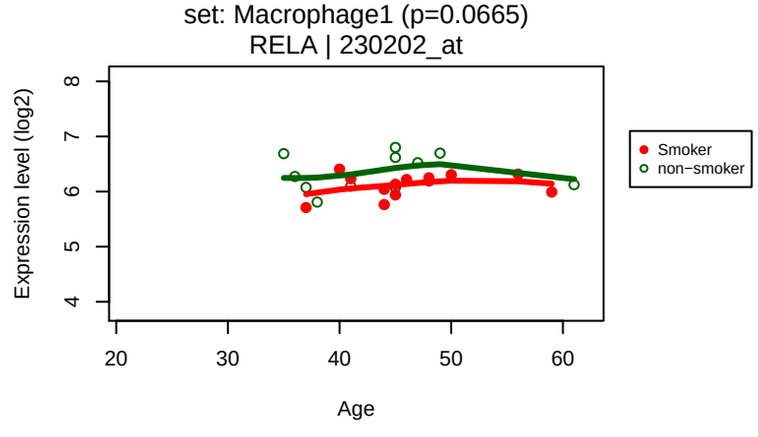

set: SmallAirway1 (p=0.1799)
RELA | 230202_at
Expression level (log2)
Age
Smoker
non−smoker

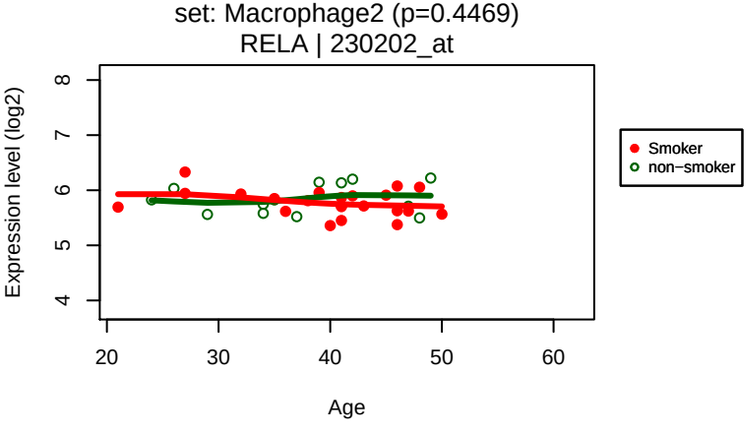

set: SmallAirway2 (p=0.0299)
RELA | 230202_at
Expression level (log2)
Age
Smoker
non−smoker

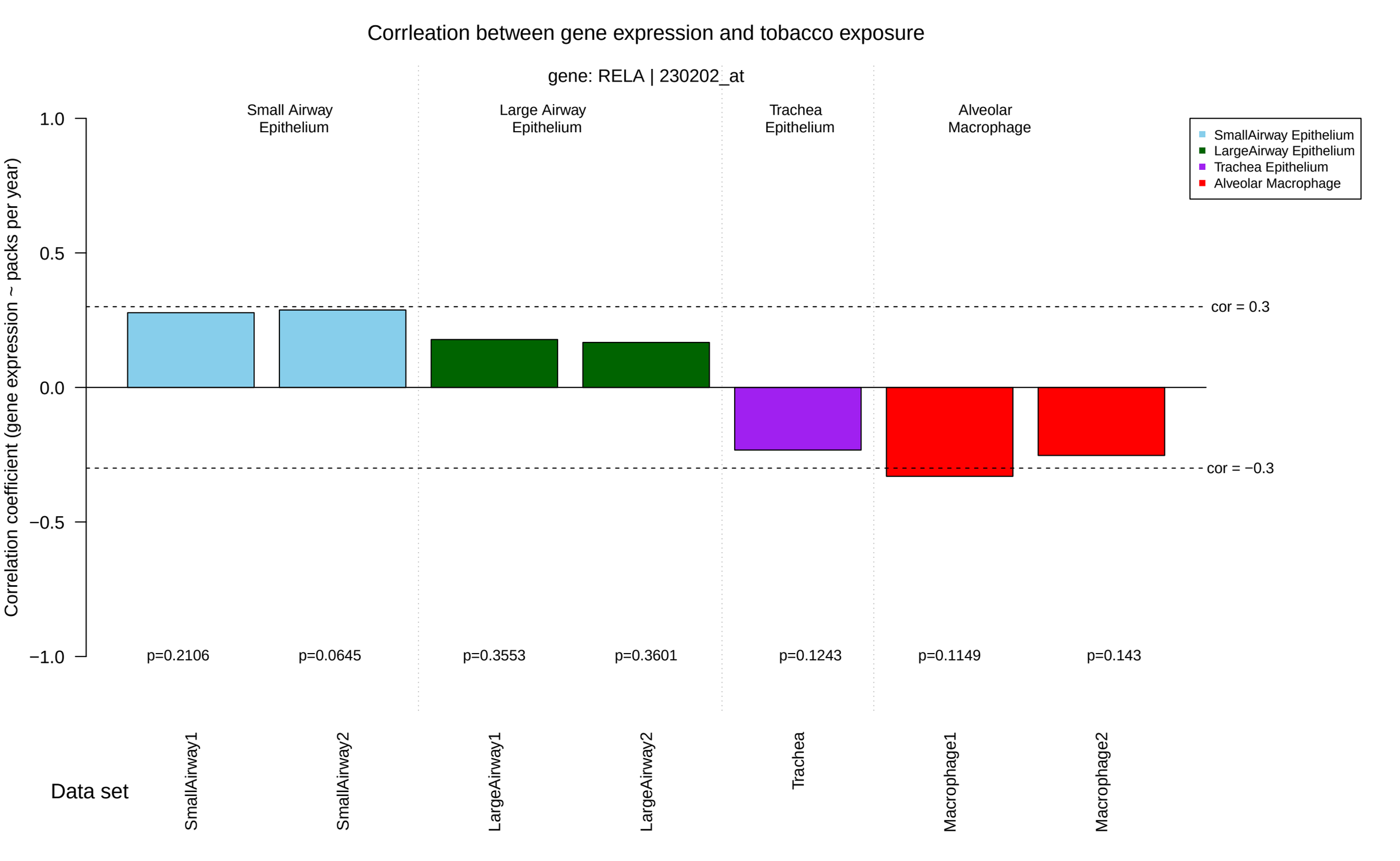

set: LargeAirway1 (p=0.1546)
RELA | 230202_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2423)
RELA | 230202_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2252)
RELA | 230202_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0665)
RELA | 230202_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4469)
RELA | 230202_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RELA | 230202_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2106
p=0.0645
p=0.3553
p=0.3601
p=0.1243
p=0.1149
p=0.143
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2